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Associations of county-level cumulative environmental quality with mortality of chronic obstructive pulmonary disease and mortality of tracheal, bronchus and lung cancers

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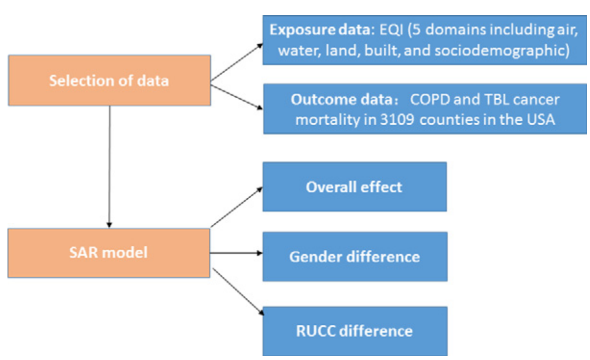
HIGHLIGHTS

- County-level EQI across US was used to capture the cumulative environmental exposures.
- The simultaneous autoregressive model was applied to assess potential spatial heterogeneity.
- Poor cumulative environmental quality increased COPD mortality and TBL cancers mortality risk.
- These associations are mainly driven by the air quality and built environment domain.
- Females appear to be more susceptible to the effect of cumulative environmental quality.

GRAPHICAL ABSTRACT



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ABSTRACT

Chronic obstructive pulmonary disease (COPD) and tracheal, bronchus, and lung (TBL) cancers are among the leading causes of mortality worldwide. Many environmental factors have been linked to COPD and TBL cancers. This study examined the associations of cumulative environmental quality indices with COPD mortality and TBL cancers mortality, respectively. Environmental Quality Index (EQI) was constructed to represent cumulative environmental quality for the overall environment and 5 major environmental domains (*e.g.*, air, water, built). Associations of each EQI indices with COPD mortality and TBL cancers mortality, across 3109 counties in the 48 contiguous states of the US, were examined using simultaneous autoregressive (SAR) models. Stratified analyses were conducted in females *versus* males and according to rural-urban continuum codes (RUCC) to assess the heterogeneity across the overall population. Overall poor environmental quality was associated with a percent difference (PD) of 0.75 [95% confidence intervals (95% CI), 0.46, 1.05] in COPD mortality and an PD of 1.22 (95% CI, 0.97, 1.46) in TBL cancers mortality. PDs were higher in females than in males for both COPD and TBL cancers. The built domain had the largest effect on COPD mortality (PD, 0.85; 95% CI, 0.58, 1.12) while the air domain had the largest effect on TBL cancers mortality (PD, 1.54; 95% CI, 1.31, 1.76). The EQI-mortality associations varied among

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different RUCCs, but no consistent trend was found. This result suggests that poor environmental quality, particularly poor air quality and built environment quality may increase the mortality risk for COPD and that for TBL cancers. Females appear to be more susceptible to the effect of cumulative environmental quality. Our findings highlight the importance of improving overall and domain-specific cumulative environmental quality in reducing COPD and TBL cancer mortalities in the United States.

1. Introduction

Chronic obstructive pulmonary disease (COPD) and tracheal, bronchus, and lung (TBL) cancers are among leading causes of morbidity and mortality worldwide. In 2017, COPD was the 7th leading cause of global years of life lost (YLLs), which reflects a 13.2% increase from the year of 2007 (GBD, 2018). Meanwhile, TBL cancers cause 1.88 million (95%UI, 1.84, 1.92) deaths throughout the world (GBD, 2018).

COPD and TBL cancers share common risk factors (Carr et al., 2018). Increasingly, many environmental factors have been linked to COPD and TBL cancers. Ambient air pollution, such as particles with a diameter ≤2.5 μm ($PM_{2.5}$) and ozone, has been associated with COPD (de Miguel-Diez et al., 2019; Lim et al., 2019) and TBL cancers (Tseng et al., 2019). The built environment has been investigated for its role in lung cancer (Bidoli et al., 2016). Sociodemographic factors (*e.g.*, low education level) have also been identified as risk factors for COPD (Wang et al., 2018). In addition, exposure to land use contaminants (*e.g.* radon, peptide) has been associated with increased risk of COPD (Yi et al., 2014) and lung cancer (Bonner et al., 2017; Liao et al., 2018; Yitshak-Sade et al., 2019). Similarly, water quality provides another potential mechanism for exposure; for example, drinking arsenic contaminated water increases lung cancer mortality in long term (Smith et al., 2018). However, in the real world, the influence of different environmental factors often occurs simultaneously. Research up to date has primarily been limited to examining single environmental exposures, few studies have explored the cumulative effect of multiple exposures across environmental domains. One of the potential reasons may be related to the challenge of quantifying the broad range of cumulative environmental exposures.

To address this limitation, the U.S. Environmental Protection Agency (EPA) developed the Environmental Quality Index (EQI). The EQI is a publicly available, county-level measure of cumulative environmental exposures across five key environmental domains (air, water, land, built, and sociodemographic) throughout the United States. The county-level EQI is the only aggregate environmental quality index available to the public, offering a complete coverage of the United States. The EQI incorporates multiple environmental components that can impact mortality due to COPD and TBL cancers.

The mortality rates due to COPD and TBL cancers varied by gender and area throughout the United States (Dwyer-Lindgren et al., 2017; Mokdad et al., 2017a). However, the underlying mechanisms remain unclear. Similarly, different environmental exposure profiles may drive the variation of environmental quality across counties in the United States. Therefore, geographically precise evaluations of the association between cumulative environmental quality and mortality of COPD and that of TBL cancers would facilitate a more complete understanding of variation, providing useful insights for reducing geographic, gender-related disparities and the health burden of COPD and TBL cancers.

In this cross-sectional study, we aim to assess effects of total and domain-specific cumulative environmental quality indices on the mortality of COPD and that of TBL cancers. Additionally, we aim to examine potential gender differences in the effects and to assess potential spatial heterogeneity in the effects across four defined levels of rural to urban area continuum.

2. Materials and methods

We included the population living in the contiguous United States, excluding Hawaii and Alaska. Hawaii and Alaska were excluded due to their geographical distances from the contiguous United States, as such long distances could introduce high uncertainty into the spatial model used in our analysis. Similarly, previous studies also excluded these two states (Fann et al., 2017; Jian et al., 2017).

2.1. Exposure data: EQI

EQI incorporates 5 domain-specific (air, water, land, built, and sociodemographic) indices into a single index to represent the cumulative environmental exposures of the US at the county-level from 2000 to 2005. A complete description of the dataset used in the EQI can be found in Lobdell et al. (2011) and the construction of the EQI has been described by Messer et al. (2014).

The air, water, and land domains were identified using the EPA's report on the environment ((EPA), 2008), while the built and sociodemographic domains were based on the literature review and consultation with scientists (Lobdell et al., 2014). Overall, 187 data sources across the five domains were evaluated for data quality, availability and geographic coverage (Lobdell et al., 2011).

Briefly, the EQI was constructed using principal component analysis (PCA) to reduce 219 environmental variables into five domain-specific (air, water, land, built, and sociodemographic) indices: air domain (87 variables, *e.g.* criteria and hazardous air pollutants), water domain (80 variables, *e.g.* overall water quality, general water contamination, recreational water quality, drinking water quality, atmospheric deposition, drought, and chemical contamination), land domain (26 variables, *e.g.* agriculture, pesticides, contaminants, facilities and radon), built environment domain (14 variables, *e.g.* primary street/highway proportion, road safety, public transit behavior, assess to various business environment including health care, and subsidized housing environment), and sociodemographic domain (12 variables, *e.g.* income, education, unemployment and poverty). All variables in each domain are presented in Supplemental Table 1. These indices were then included in a second PCA to produce an overall EQI (Lobdell et al., 2014).

To account for differences in environmental quality across rural and urban areas, the overall and domain-specific EQIs were stratified by rural-urban continuum codes (RUCC). RUCCs are assigned by the USDA and used nine-item categories to describe how metropolitan or rural an area is. Consistent with the approach used in previous studies (Gray et al., 2018a; Gray et al., 2018b; Jagai et al., 2017; Jian et al., 2017), we used 4 categories generated from the original nine-item categories, including metropolitan-urban (RUCC1), non-metropolitan urban (RUCC2), less urban (RUCC3) and thinly populated (RUCC4).

In the present study, we conducted analyses using the cumulative EQI and the domain-specific indices across the entire 48 contiguous United States. The data were retrieved from the US EPA website (https://www.epa.gov/healthresearch/epas-environmental-quality-index-supports-public-health).

2.2. Outcome data

We included age-adjusted mortality rates of COPD and TBL cancers during 2006–2010, which is subsequent in time to the EQI exposure data representing the environment quality during 2000–2005. Both COPD and TBL cancer are chronic disease; and it is reasonable to examine the cumulative effects of environmental exposure. Hence, we used a strategy similar to that used in previous studies (Huang et al., 2019;

Jagai et al., 2017) by relating outcomes with exposures assessed a few years earlier.

We calculated the overall annual average mortality rate of COPD and TBL cancer during 2006–2010. The annual mortality rates of these two diseases in each year of 2006–2010 are presented in Supplemental Fig. 1.

Age-adjusted mortality rates of COPD and those of TBL cancers for 3109 counties in the contiguous United States were obtained from the Institute for Health Metrics and Evaluation (IHME) (Mokdad et al., 2017b) (http://ghdx.healthdata.org/us-data). IHME research used de-identified death records from the National Center for Health Statistics (NCHS) and population counts from the U.S. Census Bureau, NCHS, and the Human Mortality Database and small area estimation models in order to estimate county-level mortality rates.

2.3. Covariate data

Based on literature review, we included several potential influential factors in the analysis. The county-level population density was obtained from U.S. Census Bureau (2010) (http://www.census.gov/). The percent of white population, cigarette smoking rate (overall, females and males), and alcohol consumption rate by county in 2010 were also obtained from IHME.

2.4. Statistical analyses

We used simultaneous autoregressive (SAR) models to assess the associations of EQI with COPD mortality and TBL cancers mortality, respectively. SAR models are statistical models that taking the spatial dependence into consideration (Ver Hoef et al., 2018). It's one of the most commonly used models to control the spatial autocorrelation in the residuals (Havard et al., 2009). It considers spatial dependence as an additional variable in the regression equation and estimates its effect simultaneously with that of other explanatory variables (Chakraborty et al., 2011). The base model is described as follows:

$$Log(y_i) = \rho WY + \alpha_0 + \alpha_1 \times smoking_i + \alpha_2 \times alcohol_i + \alpha_3 \times white\%_i + \alpha_4 \times popden_i + \beta \times EQI_i + \varepsilon$$

where y is the county-level age-adjusted mortality rate; ρ is the spatial autoregressive parameter; W is the spatial connectivity matrix that defined the notion of neighborhood between geographic unites, which was constructed using the queen criterion including the vertices of the lattice to define contiguities; *smoking* is the county-level cigarette smoking prevalence; *alcohol* is the county-level drinking prevalence; *white%* is the percent of the white population; *popden* is population density; *EQI* is an EQI index for each county (total EQI and different EQI domains were included in the model separately); i is the index for county; α_0–α_4 and β are model coefficients.

To address the gender difference, we used the gender-specific mortality data to evaluate the associations in male and female, respectively. We also used RUCC stratified EQI to investigate potential rural-urban differences in the associations.

In sensitivity analyses, we used the gender specific smoking prevalence instead of the overall smoking prevalence for each county, to control for the smoking difference between females and males. We also compared the main results with the associations between EQI, COPD mortality, and TBL cancers mortality during 2000 to 2005, which is in the same time period for the EQI data.

We reported the percent difference (PD), which was defined as the percentage change in annual mortality for each 1-unit increase in EQI, with 95% confidence intervals (95% CI). We used two-sided statistical testing and defined statistical significance as $P < 0.05$. All analyses were performed using the R software (version 3.6.0).

3. Results

3.1. Population description

The average annual county-level age-adjusted COPD and TBL cancers mortality rate was 53.94 (SD, 13.94) and 63.73 (SD, 16.72) cases per 100,000 population, respectively. Mortality rates were higher in the male population (COPD, 68.43 ± 17.98 per 100,000; TBL cancers, 84.49 ± 25.4 per 100,000) than in the female population (COPD, 44.25 ± 12.76 per 100,000; TBL cancers, 47.68 ± 11.68 per 100,000). Mortality rates varied across rural-urban regions. Generally, RUCC-stratified mortality rates were highest in less urbanized (RUCC3) area, followed by nonmetropolitan urbanized (RUCC2) area. However, the mortality rate was lowest in metropolitan urbanized (RUCC1) area for COPD and lowest in thinly populated (RUCC4) area for TBL cancers. The county-level age-adjusted mortality rates by RUCCs in the contiguous United States are presented in Table 1. The county-level EQI, age-adjusted mortality rates of COPD and TBL cancers are presented in Fig. 1.

3.2. Association between EQI and COPD mortality

Overall, poor environmental quality was positively associated with COPD mortality (PD, 0.75; 95% CI, 0.46, 1.05). The effect was significant for females (PD, 1.73; 95% CI, 1.37, 2.10) but marginally significant for males (PD, 0.28; 95% CI, −0.03, 0.59) (Table 2). Significant associations between total EQI and COPD were identified in metropolitan urbanized (RUCC1) (PD, 0.95; 95% CI, 0.41, 1.48) and nonmetropolitan urbanized (RUCC2) (PD, 1.79; 95% CI, 0.69, 2.89) areas, but not in less urbanized (RUCC3) (PD, −0.48; 95% CI, −1.27, 0.32) and thinly populated (RUCC4) (PD, −0.54; 95% CI, −1.62, 0.54) areas (Table 3).

In the domain-specific models, the non-stratified estimates indicate an increased COPD mortality rate associated with poor build environment quality (PD, 0.85; 95% CI, 0.58, 1.12). Increased risks of COPD mortality were associated with poor water quality (PD, 0.51; 95% CI, 0.27, 0.76) and poor air quality (PD, 0.39; 95% CI, 0.13, 0.65). In contrast, a marginal effect was identified for the sociodemographic domain (PD, 0.27; 95% CI, −0.04, 0.59), whereas no evidence was found to support a COPD or TBL mortality association with the land domain (Table 2).

In gender-stratified analyses of the domain-specific EQI, the effects of built (females, PD, 1.54; 95% CI, 1.22, 1.86 *vs* males, PD, 0.59; 95% CI, 0.31, 0.87) and water (females, PD, 0.79; 95% CI, 0.51, 1.08 *vs* males, PD, 0.27; 95% CI, 0.02, 0.52) domains were consistently significant across gender while the effects of air, land and sociodemographic domains were only significant in females but not males (Table 2).

In RUCC-stratified analyses of the domain-specific EQIs, patterns of domain specific associations varied by RUCC stratum (Table 3). In metropolitan urbanized (RUCC1), the pattern of domain-specific estimates is similar to the non-stratified result. In nonmetropolitan urbanized (RUCC2), estimates in the land (PD, 1.30; 95% CI, 0.26, 2.35) and built (PD, 2.19; 95% CI, 1.32, 3.07) domains were significant. However, in less urbanized (RUCC3) and thinly populated (RUCC4) areas, only the

Table 1
The average annual county-level age-adjusted COPD mortality rate and TBL cancer mortality rate, 2006–2010 (per 100,000 population).

Group		COPD	TBL cancers
Overall		53.94 ± 13.94	63.73 ± 16.72
Gender	Female	44.25 ± 12.76	47.68 ± 11.68
	Male	68.43 ± 17.98	84.49 ± 25.40
Area	RUCC1	51.50 ± 12.54	63.29 ± 14.64
	RUCC2	55.03 ± 12.15	64.72 ± 13.93
	RUCC3	56.43 ± 14.45	66.07 ± 18.27
	RUCC4	53.46 ± 15.36	60.25 ± 17.95

COPD: chronic obstructive pulmonary disease; TBL cancers: tracheal, bronchus, and lung cancer; RUCC, rural-urban continuum codes.

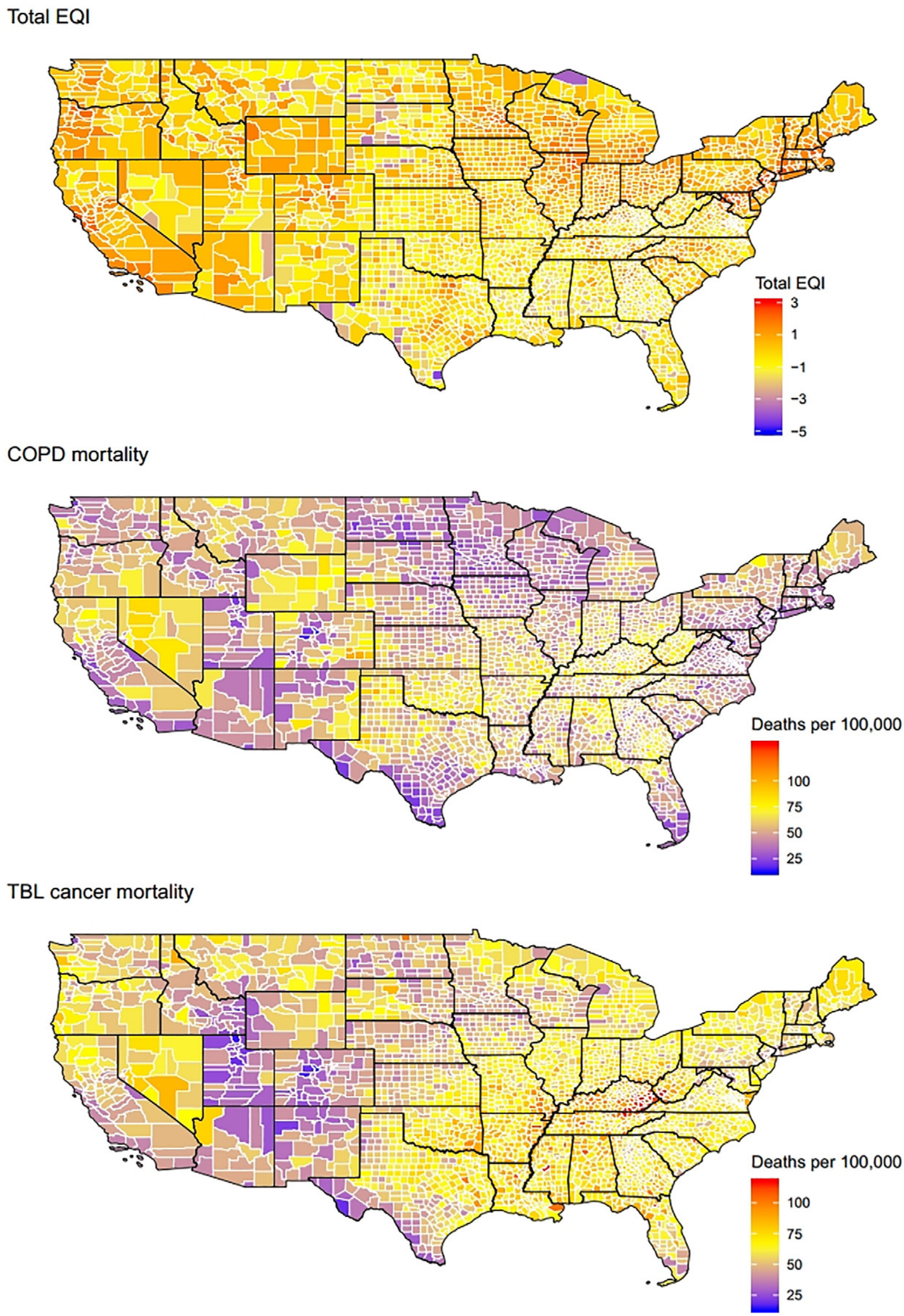

Fig. 1. The county-level EQI, age-adjusted mortality from COPD and TBL cancers.

Table 2
Associations of total and domain-specific cumulative EQI with COPD mortality and TBL cancers mortality, stratified by gender.

Population	EQI index	COPD PD (95% CI)	TBL cancer PD (95% CI)
Total (male+female)	total_EQI	0.75 (0.46, 1.05)	1.22 (0.97, 1.46)
	air_EQI	0.39 (0.13, 0.65)	1.54 (1.31, 1.76)
	water_EQI	0.51 (0.27, 0.76)	0.17 (−0.02, 0.37)
	land_EQI	−0.13 (−0.44, 0.17)	−0.24 (−0.48, 0.01)
	sociod_EQI	0.27 (−0.04, 0.59)	0.76 (0.50, 1.03)
	built_EQI	0.85 (0.58, 1.12)	0.74 (0.52, 0.96)
Female	total_EQI	1.73 (1.37, 2.10)	1.72 (1.45, 2.00)
	air_EQI	1.28 (0.96, 1.59)	1.65 (1.40, 1.91)
	water_EQI	0.79 (0.51, 1.08)	0.49 (0.28, 0.71)
	land_EQI	−0.28 (−0.64, 0.09)	−0.02 (−0.30, 0.26)
	sociod_EQI	0.92 (0.54, 1.30)	0.89 (0.59, 1.18)
	built_EQI	1.54 (1.22, 1.86)	1.25 (1.01, 1.49)
Male	total_EQI	0.28 (−0.03, 0.59)	1.22 (0.95, 1.49)
	air_EQI	−0.11 (−0.38, 0.16)	1.68 (1.43, 1.93)
	water_EQI	0.27 (0.02, 0.52)	−0.01 (−0.23, 0.21)
	land_EQI	0.1 (−0.22, 0.42)	−0.24 (−0.52, 0.04)
	sociod_EQI	−0.09 (−0.42, 0.24)	0.81 (0.52, 1.10)
	built_EQI	0.59 (0.31, 0.87)	0.72 (0.48, 0.96)

COPD: chronic obstructive pulmonary disease; TBL cancers: tracheal, bronchus, and lung cancers; EQI: environmental quality index; PD, percent difference; 95% CI: 95% confidence intervals.

estimates of built (PD, 2.19; 95% CI, 1.52, 2.87) and water (PD, 0.66; 95% CI, 0.13, 1.18) domains were positively significant, respectively.

3.3. Association between EQI and TBL cancers mortality

In general, positive association was identified for TBL cancers mortality and EQI (PD, 1.22; 95% CI, 0.97, 1.46). It's significant in both females and males, but the effect size in females (PD, 1.72; 95% CI, 1.45, 2.00) was higher than that in males (PD, 1.22; 95% CI, 0.95, 1.49) (Table 2). These associations were highest in the nonmetropolitan urbanized (RUCC2) area (PD, 1.03; 95% CI, 0.06, 2.02), followed by metropolitan urbanized (RUCC1) (PD, 0.94; 95% CI, 0.47, 1.41), but were null in the less urbanized (RUCC3) and thinly populated (RUCC4) areas (Table 3).

When considering domain-specific associations, the most significant effect was observed in the air domain (PD, 1.54; 95% CI, 1.31, 1.76) (Table 2), and the patterns of associations were similar across rural-urban strata (Table 3) and gender (Table 2), demonstrating an increase in TBL cancers mortality with poor air quality. Estimates for the sociodemographic (PD, 0.76; 95% CI, 0.50, 1.03) and built (PD, 0.74; 95% CI, 0.52, 0.96) domains were positive in overall and gender-stratified analyses, although no significant associations in the RUCC-stratified analyses. Effect estimates for the water and land domain were not significant and varied by RUCC.

3.4. Sensitivity analyses

When gender specific smoking prevalence was taken into consideration, it did not alter the overall results. For both COPD and TBL cancers, the magnitudes of PD were higher in females [COPD, 1.57 (1.22, 1.93); TBL cancers, 1.53 (1.26, 1.80)] than in males [COPD, 0.40 (0.09, 0.72); TBL cancers, 1.34 (1.06, 1.62)]. The results are presented in Supplemental Table 2.

In sensitivity analyses using COPD and TBL cancers mortality data during 2000 to 2005, similar patterns as shown in the main analyses. The results are presented in Supplemental Tables 3–4.

4. Discussion

We found that poor cumulative environmental quality was associated with both increased COPD mortality and increased TBL cancers mortality. We also found that the associations varied among different domains. Poor air quality was associated with increasing COPD mortality and TBL cancers mortality. This supports the previous estimation that about 1.6 million COPD deaths and 0.5 million lung cancer deaths can be attributed to air pollution (Schraufnagel et al., 2019). Many different pollutants, such as $PM_{2.5}$ and ozone (Cakmak et al., 2018) have irritant effects that can induce respiratory infections (Schraufnagel et al., 2019) and significantly reduced lung function (Lepeule et al., 2014), which are symptoms of COPD and could contribute to lung cancer later (Carr et al., 2018). A previous study by Jian et al. (2017) estimated the association between EQI and overall mortality (including cancer). However, they did not exam the specific effect estimates of COPD and TBL cancers. Moreover, Jian et al. used a multiple linear regression model while we applied a SAR model that took the spatial dependence into consideration. The present study adds unique insights to the literature concerning the effect of cumulative and composite environmental quality indices on COPD and TBL cancers across the entire 48 contiguous states of the United States.

Poor built environment quality was associated with increased mortality risk due to COPD and that due to TBL cancers, and the associations were significant in both females and males. Several reasons may contribute to the association. First of all, road/traffic density is included in EQI built domain. Similarly, in the Canadian Census Health and Environment (CanCHEC) Cohort, higher road density and closer proximity to major traffic roads were each associated with increased mortality risks from COPD and from lung cancer (Cakmak et al., 2019). Second, limited access to health care services may be another contributor to the exacerbation of COPD and TBL cancers mortality. For example, pulmonary rehabilitation (PR) is an effective therapy for patients with COPD, but significant geographic disparities existed in accessibility to hospital outpatient PR in the United States (Moscovice et al., 2019). Last but not least, built environment itself may impact the respiratory health

Table 3
Associations of total and domain-specific cumulative EQI indices with COPD mortality and TBL cancer mortality, stratified by rural-urban continuum codes (RUCC).

Mortality	EQI index	RUCC1 (metropolitan-urban) PD (95% CI)	RUCC2 (non-metropolitan urban) PD (95% CI)	RUCC3 (less urban) PD (95% CI)	RUCC4 (thinly populated) PD (95% CI)
COPD	total_EQI	0.95 (0.41, 1.48)	1.79 (0.69, 2.89)	−0.48 (−1.27, 0.32)	−0.54 (−1.62, 0.54)
	air_EQI	0.77 (0.28, 1.25)	0.08 (−0.93, 1.09)	−1.39 (−1.98, −0.80)	−1.96 (−3.16, −0.75)
	water_EQI	1.00 (0.57, 1.43)	0.43 (−0.44, 1.30)	0.47 (−0.08, 1.01)	0.86 (0.15, 1.58)
	land_EQI	−0.11 (−0.59, 0.36)	1.49 (0.38, 2.60)	−1.19 (−1.89, −0.49)	−1.16 (−2.20, −0.12)
	sociod_EQI	0.04 (−0.48, 0.55)	−0.24 (−1.37, 0.89)	−0.79 (−1.57, −0.01)	−0.58 (−1.77, 0.61)
	built_EQI	0.73 (0.22, 1.24)	2.4 (1.48, 3.33)	2.34 (1.64, 3.04)	0.19 (−0.59, 0.99)
TBL cancers	total_EQI	0.94 (0.47, 1.41)	1.03 (0.06, 2.02)	0.07 (−0.61, 0.74)	−1.27 (−2.16, −0.39)
	air_EQI	1.68 (1.27, 2.09)	2.19 (1.33, 3.05)	2.38 (1.90, 2.87)	3.37 (2.42, 4.33)
	water_EQI	−0.41 (−0.78, −0.03)	−0.66 (−1.43, 0.11)	−0.30 (−0.76, 0.16)	0.26 (−0.33, 0.85)
	land_EQI	−0.52 (−0.93, −0.10)	−0.25 (−1.24, 0.74)	−1.85 (−2.43, −1.27)	−2.2 (−3.04, −1.35)
	sociod_EQI	−0.82 (−1.27, −0.38)	0.47 (−0.52, 1.47)	0.08 (−0.58, 0.74)	−0.85 (−1.83, 0.12)
	built_EQI	0.20 (−0.25, 0.65)	0.56 (−0.29, 1.41)	0.40 (−0.20, 1.00)	−0.22 (−0.87, 0.42)

COPD: chronic obstructive pulmonary disease; TBL cancers: tracheal, bronchus, and lung cancers; EQI: environmental quality index; PD, percent difference; 95% CI: 95% confidence intervals.

indirectly by affecting life style and working status (Brownson et al., 2009; Wang et al., 2016).

For the sociodemographic domain, a significant effect was observed for TBL cancers, while a marginal effect (PD, 0.27; 95% CI, −0.40, 0.59) was identified for COPD mortality. Poverty is an important variable in the sociodemographic domain, which has been associated with increased risk of COPD (Raju et al., 2019) and lung cancer (Lehrer et al., 2014) in the United States. It could partially explain the associations of the mortalities and the sociodemographic domain.

For the water quality domain, we found a significant effect on COPD mortality (PD, 0.51; 95% CI, 0.27, 0.76) but a marginal effect on TBL cancers mortality (PD, 0.17; 95% CI, −0.02, 0.37). Given that arsenic concentration was one of the parameters used in the water quality domain, arsenic exposure may be related to the association we observed between COPD/TBL cancers and the water quality domain. Previous study showed that drinking arsenic contaminated water increases lung cancer mortality with a long latency period (Smith et al., 2018). And exposure to low- to moderate-dose arsenic through drinking water has been associated with impaired lung function (Parvez et al., 2013). However, there is lack of studies focusing on the effect on COPD. More studies are needed to assess the COPD and mortality effects of overall water quality.

Even though COPD and lung cancer have been associated with land contaminants such as pesticides (Bonner et al., 2017) and radon (Yitshak-Sade et al., 2019), we did not identify associations of land domain with either COPD or TBL cancers. One possible explanation for the unclear associations observed in this study was the spatial scale (Jian et al., 2017). The exposure data were at the county level; it may not be the best geographic unit for data aggregation for variables in particular environmental domains especially for those with high heterogeneity in individual exposure. For example, individual exposure to land and water contaminants may vary substantially across individuals in a county. This limitation may be a potential reason for the null associations observed in the land and water domains. Data at finer spatial scales such as the census tract level would be helpful to ascertain the results especially those seemingly counterintuitive (Jian et al., 2017).

In the present study, effect estimates varied across different rural-urban strata, with poor environmental quality associated with increasing COPD and TBL cancers in the metropolitan urbanized (RUCC1), non-metropolitan urbanized (RUCC2) areas, but not associated or inversely associated with COPD and TBL cancers in the less urbanized (RUCC3) and the thinly populated (RUCC4) areas. The model clustered by RUCC showed that among the 5 domains, air domain had the largest positive association with TBL cancers for the contiguous United States, while it only significantly increased the risk of COPD in the metropolitan urbanized areas (RUCC1). For the built domain, positive associations were observed in more populated areas with COPD (RUCC1-3), but not TBL cancers. The associations of the water, land, or sociodemographic domain with COPD and TBL cancers were more heterogeneous and no specific pattern are indented.

Considering gender, the effect estimates for associations of EQI with COPD and TBL cancers in most domains were greater in females than in males. The difference was robust in sensitivity analyses. The gender susceptibility observed in the present study appears to be consistent with a previous study finding that the effect of outdoor $PM_{2.5}$ on all-cause mortality was greater in females than in males in the entire Medicare population of the continental United States (Di et al., 2017). Gender can directly affect exposure and health-related inequalities arising from biological, social and behavioral differences. On one hand, women might be biologically more susceptible to toxic substances (Ben-Zaken Cohen et al., 2007). For example, it is known that women absorb and store environmental chemicals and metals from air, water, land in different ways compared to men (Arbuckle, 2006; Organization, 2016; Proietti et al., 2013). On the other hand, due to behavioral and occupational differences, women receive higher exposures to cooking fumes and other indoor air pollutants (WHO, 2010).

The present study has several strengths. Compared with studies using single exposures, the use of EQI was more likely to capture the health effects resulting from the overall burden of environmental exposures. It included broad environmental context and multiple domains at the county scale and covered the 48 contiguous states in United States. EQI also included variables considering the potential confounders at regional level such as income, education, unemployment, poverty. The results, even though limited by some of the domain-specific uncertainties and counterintuitive findings, suggest potential impact of environment on COPD mortality and TBL cancers mortality. Another major strength is the inclusion of spatial heterogeneity. SAR models have been increasingly used to examine spatially varying patterns and relationships in chronic conditions (Chakraborty et al., 2011; Havard et al., 2009). Comparing with linear regression model, SAR model could avoid the bias and incorrect inferences induced by spatial dependence or more formally as spatial autocorrelation (Chakraborty et al., 2011). For data that have an inherent spatial component such as mortality rate (Sparks and Sparks, 2009), SAR models separate the underlying spatial processes from the noise inherent in the data, which could not be revealed by conventional regression approaches.

Even though the majority of associations between overall EQI/EQI domains and COPD/TBL cancers were positive, several negative associations were still observed for the water, land, and sociodemographic indices among subgroup analyses on RUCCs. A possible explanation was that the county-level assessment might not be ideal for all the domains in EQI. Variables were evaluated based on their overall contributions to a given domain across the United States, which occurred prior to stratification by RUCC (Gray et al., 2018a). If the variables within a domain are more heterogeneous at the county level, the lack of representativeness could introduce exposure misclassification which might contribute to the potentially lower effect estimates in more urbanized areas where more heterogeneity is expected (Jian et al., 2017).

Several limitations should also be noted. First, this is an ecological cross-sectional study. The ecological nature does not allow us to evaluate individual risk factors for COPD and TBL cancers, such as personal smoking exposure, and personal exposure of air pollution. However, we controlled overall and gender specific smoking rates at the county level in statistical analyses. Second, information on noise, green land, indoor air pollution (Balmes, 2019) and healthcare (insured *vs* non-insured) were not available. Therefore, even though the majority of associations between overall EQI/EQI domains and COPD/TBL cancers were positive we are not able to evaluate the potential effects. Third, the EQI measures were comprehensive but abstract, making it impossible to differentiate the influence of each specific contributing factor. In addition, due to the lack of accessibility to the county-level covariates (*e.g.* smoking rates) during the same time period as EQI (2000–2005), we have to use covariates in 2010, the earliest year when the data became available. Upon the availability of more recent EQI data, future studies may adopt a longitudinal study design and examine the change of mortality pattern in response to the change of environmental quality throughout the United States. Upon the availability of more recent EQI data, future studies may adopt a longitudinal study design and examine the change of mortality pattern in response to the change of environmental quality throughout the United States. Another limitation of the present study is the lack of consideration of potential confounding from climate, as the climate conditions varied across the counties. We deem that the climate impact requires an in-depth analysis especially considering different counties may have experienced different degrees of climate changes. We recommend future studies to examine the climate impact independently or in combination of EQIs.

Overall, our study provides useful information on how different aspects of the environment operate together to influence COPD and TBL cancers mortality. It also emphasizes that prioritized efforts should be made on air and built environment because it may be particularly important to control COPD and TBL cancers. However, due to the spatial heterogeneity, for policy makers, customized policy interventions that address specific and most concerning environmental issue in a local area could be more effective than a nationwide universal intervention in the future. In addition, gender-specific interventions should be implemented to address the disproportionately high effects of environment quality on mortality of COPD and that of TBL cancers for women.

5. Conclusions

In this analysis of county-level data across the 48 contiguous United States, we found that poor cumulative environmental quality is associated with increased COPD mortality and TBL cancers mortality, respectively. These associations are mainly driven by the air quality composite index and the built environment composite index. We also found that the magnitude of association (effect size) is greater in females than in males. Our findings highlight the importance of cumulative environmental exposures on health outcomes as reflected in two leading causes of mortality (COPD and TBL cancers). The results provide evidence to support the consideration of gender and environmental domains in developing strategies for reducing COPD and TBL cancer mortalities.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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